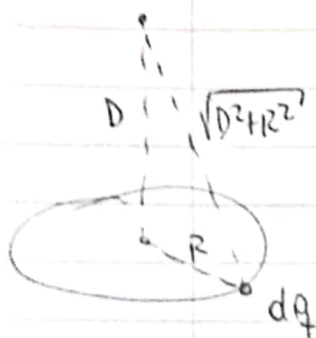
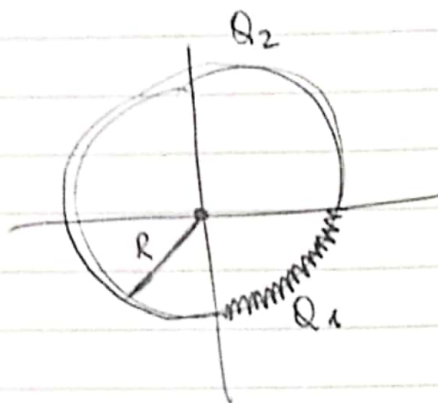


$$\lambda = dq/ds$$

(10)



$$dV = k \frac{dq}{\sqrt{D^2 + R^2}}$$

$$ds = R d\theta$$

$$dV = k \frac{\lambda ds}{\sqrt{D^2 + R^2}}$$

$$0 \leq \theta \leq \frac{3\pi}{2}$$

$$\lambda_2 = \frac{-48 \cdot 10^{-9}}{0,12 \cdot \frac{3\pi}{2}}$$

$$\lambda_i = \frac{Q_i}{s_i}$$

$$s_i = R\theta_i$$

$$\lambda_2 = -8,488 \cdot 10^{-8} \frac{C}{m}$$

$$\frac{3\pi}{2} \leq \theta \leq 2\pi$$

$$\lambda_1 = \frac{8 \cdot 10^{-9}}{0,12 \cdot \pi/2}$$

$$\lambda_1 = 4,244 \cdot 10^{-8} \frac{C}{m}$$

$$\Rightarrow V = k \left[\int_0^{3\pi/2} \frac{\lambda_2 R d\theta}{\sqrt{D^2 + R^2}} + \int_{3\pi/2}^{2\pi} \frac{\lambda_1 R d\theta}{\sqrt{D^2 + R^2}} \right]$$

$$V = \frac{k \lambda_2 R}{\sqrt{D^2 + R^2}} \cdot \frac{3\pi}{2} + \frac{k \lambda_1 R \cdot \frac{\pi}{2}}{\sqrt{D^2 + R^2}}$$

$$V = \frac{k Q_2}{\sqrt{D^2 + R^2}} + \frac{k Q_1}{\sqrt{D^2 + R^2}}$$

centro $D = 0$

$$\Rightarrow V = \frac{9 \cdot 10^9}{0,12} [8 \cdot 10^{-9} - 48 \cdot 10^{-9}]$$
$$= -3000 \text{ [V]}.$$

en el eje $D = 0,08 \text{ m}$

$$V = \frac{9 \cdot 10^9}{\sqrt{0,12^2 + 0,08^2}} [8 \cdot 10^{-9} - 48 \cdot 10^{-9}]$$

$$V = -2496,15 \text{ [V]}.$$